

REST API Basics

A short summary

REST (Representational State Transfer) is a style for building web APIs. A client asks a server for a resource, and the server returns a representation of it — usually as JSON. It is stateless: every request carries everything the server needs, and the server keeps no session memory between calls.

Core idea

- Everything is a resource, identified by a URL (e.g. `/users/42`).
- You act on resources using HTTP methods, not custom verbs in the URL.
- The server returns a status code plus a body (usually JSON).
- Each request is independent — the server does not remember previous ones.

HTTP methods (verbs)

- GET — read a resource, never changes anything.
- POST — create a new resource.
- PUT / PATCH — update an existing resource (PUT replaces, PATCH edits fields).
- DELETE — remove a resource.

Common status codes

- 200 OK — request succeeded.
- 201 Created — a new resource was created.
- 400 Bad Request — the client sent something invalid.
- 401 / 403 — not authenticated / not allowed.
- 404 Not Found — the resource does not exist.
- 500 Server Error — something broke on the server.

Good habits

- Use nouns in URLs (`/orders`), not verbs (`/getOrders`).
- Keep responses consistent and predictable.
- Use the right status code so clients can react correctly.
- Version your API (e.g. `/v1/`) so you can change it later safely.